Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/12/2007
PI Name:	Sudarshan, Tirumalai Ph.D.		
Project Title:	Novel Radiation Shielding and Structurally Efficient Materials for Space Missions		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22031	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:			
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Task Description:	Due to the harmful effects of space radiation on the human body, the development of better radiation shielding materials will be an enabling technology for long-duration space missions. The 'Specific Aim' of this proposal is to develop multifunctional materials that will adequately shield the crew from radiation danger, while serving as vehicle structure that can be load bearing and simultaneously serve as a impact shield against micro meteorites. The effectiveness of polymers as radiation shielding materials for use in long duration space missions is largely a function of their hydrogen content. The thrust of this proposal is to modify existing thermoset resins to increase their hydrogen content. The combination of new, higher hydrogen content thermosetting resins and UHMWPE fibers (with potential use of nanoparticles) will provide not only an effective shielding material but also provide the designer much greater latitude and flexibility in the design of these structures. The proposed work will focus on developing high hydrogen content, easily processed epoxy resin systems for use as spacecraft composite matrices. The first part of the research will consist of fundamental resin development and radiation testing which will be followed by composite development. Resin formulations that can be used as matrices in prepregs will be developed as well as other resin types that can be processed by low cost manufacturing methods including VARTM, RTM, and filament winding. Matrix and composite properties can and will be tailored to suit the needs of the particular application. This research will elevate epoxy resins chemistry to a new level of performance for radiation shielding while also providing the known epoxy attributes of excellent processing and mechanical properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Four new thermoset composite matrices were developed that, compared to currently available matrices, exhibited improved space radiation shielding while retaining good structural properties and excellent processability. This was done by dramatically increasing the hydrogen content of epoxy thermoset matrices from about 7 percent for commercial epoxies to about 11 percent for our new epoxy compositions of interest. In terms of the density of hydrogen atoms per cubic centimeter (cm3), the increase was from 5 to almost 8 x 10(22) atoms/cm3. These numbers rival those calculated for PE which has the highest hydrogen content relative to its density of any existing polymer (ca. 14.3 percent) and a very high hydrogen atom density, ca. 8.3 x 10(22) atoms/cm3. The tensile moduli of these new high hydrogen content epoxies are comparable to those of commercially available epoxies. Radiation shielding effectiveness analyses (via simulated radiation shielding transport property plots of dose equivalent in cSc/yr vs. depth in gm/cm2) showed that these new matrices were superior to commercial epoxies in shielding GCR and SEP radiation and are nearly as efficient as polyethylene, "the gold standard," in their radiation shielding effectiveness. Unitape prepreg was commercially made from both plasma surface-treated and untreated Spectra™ 1000 x 1300 denier fiber using one of our 250°F curing epoxies, UN-2. Composites were fabricated from both prepregs using simulated autoclave procedures and tensile, shear, and flexue data obtained. [Contractural delays in the execution of this portion of the program require an addendum to this Final Report be issued when the test data become available.] The theoretical tensile moduli of these composites (using laminate plate theory) were similar to those from Spectra™ composites made with commercial epoxies. Radiation shielding effectiveness analyses indicated that the radiation shielding properties of Spectra™ composites made with our new epoxies were better than those of Spectra™ composites made with commercial epoxies and rival those of PE/PE composites. The optional supplemental approach, fabrication of composites using our new high hydrogen epoxies and carbon fiber, was successfully executed. Prepregs were commercially made from IM-7 carbon fiber and both a 250°F cure resin, UN-2, and a 350°F cure resin, UN-3. Composites were fabricated from both prepregs using autoclave procedures. Flexural, shear, and tensile data were obtained. Their experimental tensile moduli (from laminate plate theory) were, on average, about 20-35 percent lower than the theoretical values. Radiation shielding analyses plots indicated that their radiation shielding potential, as expected, was better than IM-7 composites made with commercial epoxies but lower than composites made with Spectra™ fiber. Vacuum Assisted Resin Transfer Molding (VARTM), a standard industry technology, was demonstrated as a facile process for fabricating Spectra™ fabric composites using one of our new high hydrogen epoxies, CS-87-2, and Spectra fabric. Further experiments with this technique are required in order to produce quality laminates.
Bibliography Type:	Description: (Last Updated:)